economy immediately. Thus, he focused on the short-run. He reasoned that an increase in aggregate expenditure (which represents aggregate demand) leads to an increase in output; it does not lead to rise in prices till full employment is reached. In view of above, price level is kept fixed in the simple Keynesian model. An implication of fixed prices is that nominal values and real values are the same.

In this Unit we deal with a closed economy; there is no external trade. We begin with the derivation of IS and LM curves. Subsequently, we derive the AD curve from the equilibrium points of the IS-LM model.

3.2 EQUILIBRIUM IN THE REAL SECTOR

The aggregate expenditure (or, aggregate demand) of an economy is given by

Y = C + I + G

where Y is output (or, aggregate supply), C is consumption expenditure, I is investment expenditure, and G is government expenditure.

The consumption function is given by

$$C = a + b(Y - T) = a + bY_d \qquad \dots (3.1)$$

where Y_d is disposable income (i.e., Y – T). Here T is a lump sum tax levied by the government. The investment function is given by

$$I = c + dY - e\,i \qquad \dots (3.2)$$

where i is the rate of interest (there is no difference between nominal rate of interest and real rate of interest as price is kept fixed). Let us assume that government expenditure (G) is constant, and exogenously given. Thus,

$$Y = a + b(Y - \bar{T}) + (c + dY - e\,i) + \bar{G} \qquad \dots (3.3)$$

On re-arrangement of terms in equation (3.3), we obtain

$$Y - bY - dY = a - b\bar{T} + c - ei + \bar{G}$$

Or, $Y = \frac{a+c+\bar{G}-b\bar{T}}{(1-b-d)} - \frac{e}{(1-b-d)}i$ … (3.4)

The IS equation, in its general form, is given by

$$Y = \alpha_G(\bar{A} - ei) \qquad \dots (3.4a)$$

where $\alpha_G = \frac{1}{(1-b-d)}$ and $\bar{A} = a + c + \bar{G} - b\bar{T}$.

In equation (3.4) we find an inverse relationship between Y and i. It implies that, as i increases, there is a decrease in Y. In order to find out the slope of equation (3.4) you have to express it in terms i and find out the slope $\frac{di}{dY}$. You can find out the slope of the IS curve as $-\frac{(1-b-d)}{e}$. Depending upon the value of the parameters, the IS curve would be flatter or steeper.

3.2.1 Derivation of the IS Curve

In Fig. 3.1 we explain the relationship between Y and i through a diagram. In Panel (a) of Fig. 3.1 we take aggregate expenditure (AE) on the y-axis and output (Y) on the x-axis. Let us assume that the economy is operating with AE_0 while the rate of interest is i_0. The intersection of AE_0 with the 45^0 line at point 'a' is of utmost importance to us. Note that the 45^0 line in Fig. 3.1(a) indicates equality between aggregate demand (AE) and aggregate supply (Y), such that it corresponds to equilibrium in the economy. We plot this point 'a' in panel (b) of Fig. 3.1. In panel (b) we take interest rate (i) on the y-axis and output (Y) on the x-axis.

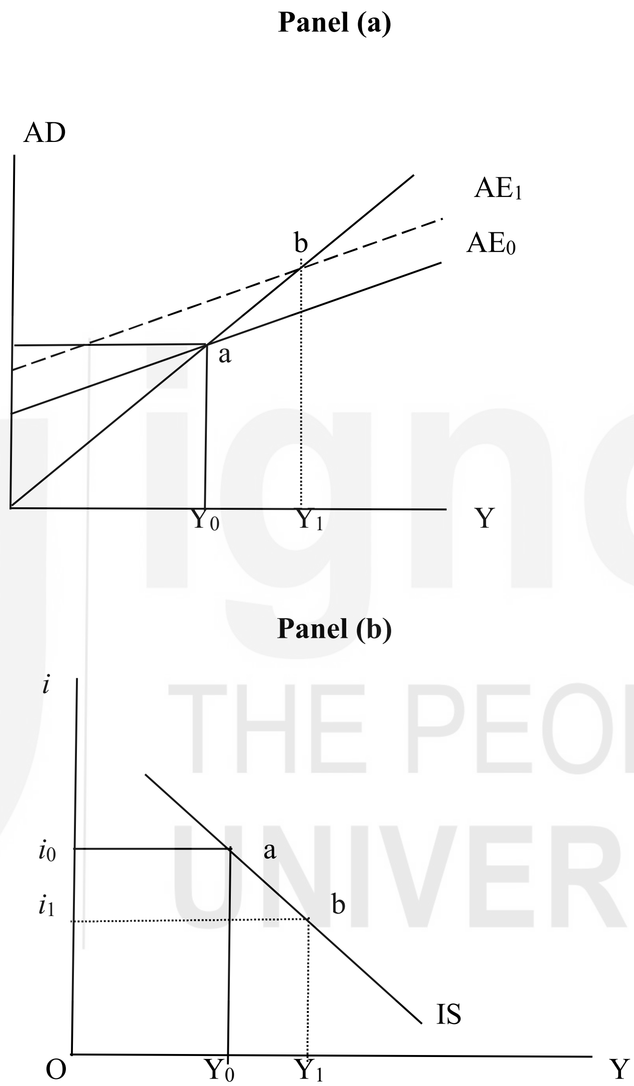

Fig. 3.1: Derivation of the IS Curve

Suppose, there is a decrease in the rate of interest from i_0 to i_1. As a result of this, the level of investment in the economy increases and the new aggregate expenditure level rises to AE_1. The aggregate expenditure curve (AE_1) intersects the 45^0 line at point-b. Due to the increase in the aggregate expenditure, there is an increase in the equilibrium level of output from Y_0 to Y_1. We plot point-b in Fig. 3.1(b) corresponding to output level Y_1 and intertest rate i_1. When we combine points 'a' and 'b' we obtain a downward-sloping curve, called the IS curve. A point on the IS curve denotes equilibrium in the real sector (also called the goods sector) of the economy and there is equality between investment (I) and saving (S). Note that the real sector of the economy is at equilibrium on each and every point of the

IS curve. An implication of the above is that any point outside the IS curve shows disequilibrium in the real sector. In Fig. 3.2 we consider two such points 'c' and 'd' which are not on the IS curve, and find out their implications.

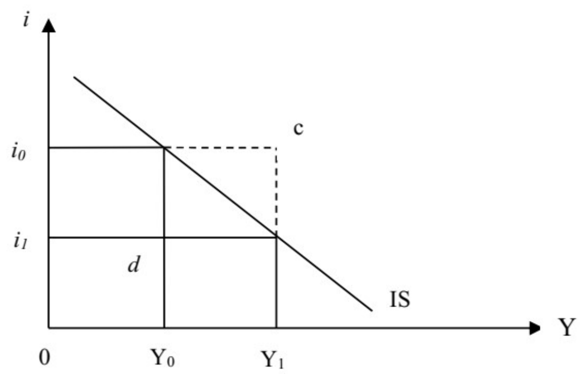

Fig. 3.2: Disequilibrium in the Real Sector

At point 'c' in Fig. 3.2, interest rate is i_0 while output is Y_1 . Notice that at Y_1 level of output, the equilibrium rate of interest should be i_1. Thus, the actual rate interest is higher than what it should. As a result, the aggregate demand will decrease (largely due to decrease in investment) and the rate of interest will decline. It implies that at point-c, there is excess supply of goods in the market. We can generalize that any point which is above and to the right of the IS curve, indicates excess supply of goods.

Let us look at point 'd' which is below and to the left of the IS curve. At point 'd' in Fig. 3.2, income is at Y_0 but the rate of interest is at i_1. Since the rate of interest is lower than the equilibrium rate of interest, the demand for investment will increase. Consequently, the rate of interest will rise upward. Thus, at point 'd' there is excess demand for goods. We can generalize that at any point below and to the left of the IS curve, there is excess demand for goods.

3.2.2 Shift in the IS Curve

The IS curve depicts the relationship between income and interest rate. Therefore, any change in other parameters or variables will result in a shift in the IS curve. The position and the slope of the IS curve depends upon the value of the parameters in equation (3.4). If the IS curve is flatter, the economy is more sensitive to interest rate – a small decrease in interest rate will lead to a large increase in income. On the other hand, a steeper IS curve implies insensitivity to interest rate change.

There are two exogenous variables in equation (3.4), viz., G and T. In equation (3.4) we assumed that government expenditure and taxes are fixed. Thus, changes in taxes and government expenditure will also result in shifts in the IS curve. Let us look into equation (3.4). An increase in the value of $\bar{G}$ will result in an increase in the intercept; thus, there will be an upward parallel shift in the IS curve. An increase in $\bar{T}$, on the other hand, will decrease the intercept and shift the IS curve downward to the left.